

COURSE DESCRIPTION FORM

INSTITUTION National University of Computer and Emerging Sciences

PROGRAM (S) TO BE Computer Science

EVALUATED

A. Course Description

(Fill out the following table for each course in your computer science curriculum. A filled-out form should not be more than 2-3 pages.)

Course Code	CS2006																			
Course Title	Operating Systems																			
Credit Hours	3+1																			
Prerequisites by Course(s) and Topics	Data Structures																			
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	<table border="1"> <tr> <th colspan="3">Assessment with the weight.</th> </tr> <tr> <th>Assessment Type</th><th colspan="2">Weight</th> </tr> <tr> <td>Assignments</td><td colspan="2">10</td> </tr> <tr> <td>Quiz</td><td colspan="2">15</td> </tr> <tr> <td>Mid-Terms</td><td colspan="2">30</td> </tr> <tr> <td>Final</td><td colspan="2">45</td> </tr> </table>		Assessment with the weight.			Assessment Type	Weight		Assignments	10		Quiz	15		Mid-Terms	30		Final	45	
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Course Coordinator	Dr. Rana Asif Rehman																			
URL (if any)																				
Course Goals	1. Understanding components of an operating systems 2. Learn how the Operating systems evolved 3. Learning thread programming and memory management 4. Understanding the problems that are faced in process execution when shared resources are involved 5. Develop understanding of organization and structure of file system 6. Providing the knowledge for algorithms and data structures used in development of an OS.																			
Textbook (or Laboratory Manual for Laboratory Courses)	Operating Systems (10th Edition) by Abraham Silberschatz et al.																			
Reference Material	Modern Operating System, Author (s): Tenenbaum Operating Systems, Author (s): William Stallings																			

Program Learning Outcomes (PLOs):	This course covers the following PLOs: <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 10%;">PLO#</th> <th style="width: 35%;">PLO Name</th> <th style="width: 55%;">PLO Description</th> </tr> </thead> <tbody> <tr> <td>PLO 2</td> <td>Knowledge for Solving Computing Problems</td> <td>Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements</td> </tr> <tr> <td>PLO 3</td> <td>Problem Analysis</td> <td>Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines</td> </tr> <tr> <td>PLO 4</td> <td>Design/ Development of Solutions</td> <td>Understand and commit to professional ethics, responsibilities, and norms of professional computing practice</td> </tr> <tr> <td>PLO 5</td> <td>Modern Tool Usage</td> <td>Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations</td> </tr> </tbody> </table>	PLO#	PLO Name	PLO Description	PLO 2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements	PLO 3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines	PLO 4	Design/ Development of Solutions	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice	PLO 5	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations									
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Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Timeline	Content Covered
	Lecture 1	Course Introduction - Introducing syllabus, policies, and projects. - An overview of basic of OS, Kernel - Early Systems without OS
	Lecture 2	Introduction - OS booting Process - Device controller - Device drivers - Device Status Table
	Lecture 3	Interrupts - Interrupt and its types (hardware, software, traps) - Simple Interrupt processing - Synchronous, Asynchronous and DMA - Storage device hierarchy
	Lecture 4	Systems Calls - OS provided services - Systems calls and its parameters passing OS Structure - Monolithic - Layered - Mircokernal - Kernel Modules - Virtual Machines (type 1 and 2 hypervisors) - Containers (dockers)
	Lecture 5	Process Management - Process, Process address space - PCB - Context switching - Process states (2 and 5 states models)
	Lecture 6	Operations on Processes - Process creation (i.e. fork system call) - Wait system call - Zombie and orphan processes - exit, abort and exec system calls
	Lecture 7	Inter-Process Communications (IPC) - Independent and cooperating processes - IPC with shared memory

		• IPC with Message passing (direct/indirect, blocking/non-blocking, buffering) • Sockets • Remote Procedure calls (RPC) • Pipes (ordinary vs named pipes)	
	Lecture 8	CPU Scheduling • Schedulers (short-term, long-term and medium-term) • Preemptive and non-preemptive scheduling • Scheduling Criteria (CPU utilization, throughput, turnaround time, waiting time, response time) • Scheduling Algorithms (i.e. FCFS)	
	Lecture 9	CPU Scheduling-II • Scheduling Algorithms (SJF, Priority-based, Round Robin, Multi-level queue, Multi-level Feedback queue)	
	Lecture 10	Revision	
	First Mid-term Exam		
	Lecture 11	Threads • Introduction • Light-weight vs. Heavy-weight process • Single vs. Multithreaded processes • Multicore Programming • Concurrency vs. Parallelism • Hardware Threads	
	Lecture 12	Threads-II • User-level vs. Kernel level threads • Multithreading Models (Many-to-One, One-to-One, Many-to-Many) • System Calls --- pthread_create(), pthread_join(), pthread_exit(), pthread_self() • Coding examples related to threads	
	Lecture 13	Process Synchronization • Introduction • Why synchronization? • Synchronization Issues • Race Condition • Critical Section (CS)	

		Solution of CS problem (mutual exclusion, progress, bounded waiting)	
	Lecture 14	Process Synchronization-II - Implementing Mutual Exclusion (Disabling Interrupts, Lock Variables, Strict Alteration) - Peterson's solution and its issue - Bakery Algorithm	
	Lecture 15	Process Synchronization-III - Semaphores (binary vs. counting) - wait () and signal () functions - Semaphores implementation without busy waiting - Applications of Semaphores - Problems with Semaphores	
	Lecture 16	Process Synchronization-IV - Classical problems of Semaphores - Bounded-Buffer problem (consumer vs. producer) - Readers-Writer problem (reader's precedence, writer's precedence) - The Dining Philosophers problem	
	Lecture 17	Deadlock - The deadlock problem - Deadlock Characteristics (Mutual exclusion, Hold and wait, No preemption, Circular wait) - Resource Allocation Graph - Methods for Handling Deadlocks	
	Lecture 18	Deadlock-II - Deadlock Prevention - Deadlock Avoidance (safe state, unsafe state) - Resource Allocation Graph (for single instance) - Banker's Algorithm (for multiple instances)	
	Lecture 19	Deadlock-III - Deadlock Detection - Wait-For Graph (for single instance) - Detection's Algorithm (for multiple instances) - Deadlock Recovery (process termination, resource preemption)	
	Lecture 20	Revision	

Second Mid-term Exam	
Lecture 21	Memory Management • Background • Requirements: Relocation, Protection • Hardware Address Protection • Memory Management Unit (MMU)
Lecture 22	Memory Management-II • Logical vs. Physical Address space • Address binding • Static vs. dynamic loading • Static vs. dynamic linking • Memory Management (No OS vs. Uni-programming vs. Multiprogramming)
Lecture 23	Memory Management-III • Memory Allocation Mechanism (Contiguous vs. Non-contiguous) • Contiguous (Fixed vs. Dynamic Partitioning) • Non-contiguous (Fixed vs. Dynamic Partitioning)
Lecture 24	Memory Management-IV • Internal Fragmentation vs. External Fragmentation • Swapping • Compaction mechanism • Memory allocation for Processes • Memory Tracking using Bitmaps • Memory Tracking using Linked List (First-Fit, Next-Fit, Best-Fit, Worst-Fit)
Lecture 25	Memory Management-V • Paging Mechanism • Paging Hardware (address translation scheme) • Page Table • Paging Hardware with TLB • Valid and Invalid bit in a page table • Shared Pages

	Lecture 26	Memory Management-VI - Structures of Page Tables - Hierarchical Page Tables - Hashed Page Tables - Inverted page Tables		
	Lecture 27	Memory Management-VII - Numerical Questions related to Memory Management - Find Page table size and Memory size - Find Effective Access Time (EAT) - Find Number of Frames and pages - Find Logical Address to corresponding Physical Address		
	Lecture 28	Virtual Memory - Background - Virtual Address Space - Demand paging - Handling Page Fault - Free-Frame List - Page Replacement		
	Lecture 29	Virtual Memory-II - Page Replacement Algorithms - First-In First-Out (FIFO) - Optimal - Least Recently Used (LRU) - Clock (second-chance algorithm)		
	Lecture 30	Revision & Project Evaluations		
	Final Examination			
Laboratory Projects/Experiments Done in the Course				
Programming Assignments	A programming assignment where students are expected to develop an application with a focus on identifying vulnerabilities and implementing mechanisms to address them.			
Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	40	25	25	10
Oral and Written	Every student is required to submit at least 2 written reports for the given assignments			



Communications	and to make __1__ oral presentations of typically __10__ minute's duration for the project. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.
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Instructor Name: Dr. Rana Asif Rehman

Instructor Signature